

MKT-03-REGULATORY-LANDSCAPE

MKT-03: Regulatory Landscape Analysis

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EMP/CME Protection Regulatory Framework — Mandatory Demand Generation

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EXECUTIVE SUMMARY

Regulatory developments are the single most powerful demand driver for CSM’s products. Unlike consumer products that must create demand through marketing, EMP/CME hardening is increasingly mandated by regulation, creating non-discretionary demand that customers must satisfy regardless of economic conditions. This document catalogs the regulatory frameworks — existing, pending, and anticipated — that create mandatory or de facto mandatory demand for dielectric protection products.

The regulatory landscape spans four domains: (1) electric grid reliability regulation (FERC/NERC), (2) critical infrastructure protection (DHS/CISA), (3) space weather monitoring and warning (NOAA/DoD), and (4) insurance regulatory frameworks (NAIC/state insurance commissioners). Each domain is evolving toward stricter EMP/CME hardening requirements. CSM’s strategy is to engage early in regulatory proceedings, participate in standards development, and position its technology as the compliance pathway of choice.

The most consequential pending action is FERC’s anticipated Notice of Proposed Rulemaking (NOPR) on geomagnetic disturbance (GMD) standards, which would extend existing GMD planning requirements to include mandatory hardening of critical substations. If finalized as expected in 2027-2028, this would create binding demand for grid hardening products covering approximately 2,847 critical substations identified by NERC’s 2024 Long-Term Reliability Assessment.

SECTION 1: FERC/NERC GRID RELIABILITY STANDARDS

1.1 FERC Authority and Jurisdiction

The Federal Energy Regulatory Commission (FERC) regulates the interstate transmission of electricity, natural gas, and oil. Under the Energy Policy Act of 2005, FERC designated the North American Electric Reliability Corporation (NERC) as the Electric Reliability Organization (ERO) with authority to develop and enforce mandatory reliability standards for the bulk power system.

FERC Order 779 (2013) directed NERC to develop geomagnetic disturbance (GMD) reliability standards. This initiated a cascade of regulatory requirements that continues to expand.

1.2 Existing NERC GMD Standards

EOP-010-1 (Geomagnetic Disturbance Operations)

Effective: January 2024 Scope: All reliability coordinators and transmission operators with bulk power system assets above 200 kV. Requirements: - Develop and implement GMD operating plans - Monitor space weather conditions - Coordinate GMD response actions across interconnected systems - Conduct GMD operating drills and exercises

Impact on CSM: Creates awareness and operational readiness. Does not yet mandate hardware hardening, but establishes the operational framework that hardening equipment would operate within.

TPL-007-4 (Transmission System Planned Performance for GMD Events)

Effective: April 2023 (updated January 2026) Scope: All transmission planners and planning coordinators.

Requirements: - Conduct GMD vulnerability assessments every 5 years - Assess impact of a benchmark GMD event (100-year geomagnetic disturbance) - Identify transformers at risk of thermal damage from geomagnetically induced currents (GIC) - Develop corrective action plans for vulnerable assets

Impact on CSM: This standard generates the vulnerability assessments that identify specific assets requiring hardening. CSM’s products address the “corrective action plan” requirement by eliminating GIC pathways through dielectric substitution. **CSM should target utilities with the highest GIC vulnerability scores from their TPL-007 assessments.**

CIP-014 (Physical Security) — GMD Extension Anticipated

The NERC Critical Infrastructure Protection (CIP) standards address physical and cyber security. An extension to include GMD/EMP hardening for critical CIP-identified substations is under industry discussion, with a standards authorization request (SAR) expected in 2027.

1.3 Pending and Anticipated FERC/NERC Actions

FERC GMD NOPR (Expected 2027-2028)

The most consequential regulatory action for CSM is FERC’s anticipated Notice of Proposed Rulemaking on comprehensive GMD standards. Based on FERC’s 2024 technical conference on GMD, the NOPR is expected to include:

- Mandatory GIC blocking or mitigation at all bulk power system substations with high-side voltage ≥ 200 kV (approximately 4,700 substations in the US)
- Required hardening of control buildings and communication systems at critical substations
- GMD monitoring equipment at all reliability coordinator control centers
- Annual GMD readiness certification

Estimated Economic Impact: Mandatory hardening of 4,700 substations at an average cost of \$2.5 million per substation = **\$11.75 billion in mandated spending**. CSM’s dielectric approach targets 15-20% of this market, representing \$1.76-2.35 billion in potential revenue.

NERC 2024 Long-Term Reliability Assessment — GMD Findings

- Identified 2,847 critical substations where GMD-induced transformer damage would cause cascading outages affecting >500,000 customers each
- Recommended mandatory GIC mitigation for all critical substations within 5 years
- Estimated cost of \$1.2-3.8 million per substation for GMD protection (EPRI estimate, referenced in NERC report)

NERC GMD Task Force — 2026 Work Plan

- Developing performance requirements for GIC blocking devices
- Establishing testing and certification standards for GMD protection equipment
- Creating a GMD event severity scale for operational response
- Evaluating long-term hardening alternatives including dielectric and superconducting technologies

CSM Engagement Strategy: Participate in NERC GMD Task Force proceedings. Submit technical comments on dielectric protection performance characteristics. Seek inclusion of dielectric materials as an accepted compliance technology in NERC guidance documents.

1.4 State-Level GMD/EMP Initiatives

Several states have enacted EMP/CME preparedness legislation independent of federal action:

State	Legislation	Year	Requirements
Texas	SB 3 (87th Legislature)	2021	Grid weatherization and resilience; PUC directed to study EMP/CME risks
Florida	HB 5001 (appropriations)	2024	\$25M for grid hardening study including EMP/CME
			State EMP commission

Maine	LD 104 (EMP Commission)	2019	established; biennial report to legislature
Virginia	HB 2039	2017	Joint subcommittee on energy security and EMP
South Carolina	H 3354	2020	Electric utility EMP preparedness study
Louisiana	HCR 72	2021	EMP and cybersecurity task force
Arizona	SB 1476	2022	Critical infrastructure resilience including EMP

These state-level actions create patchwork demand that favors CSM’s ability to execute state-by-state deployment while federal standards develop.

SECTION 2: DHS/CISA CRITICAL INFRASTRUCTURE PROTECTION

2.1 DHS EMP/GMD Program

The Department of Homeland Security’s Cybersecurity and Infrastructure Security Agency (CISA) is the lead federal agency for critical infrastructure protection. Under Executive Order 13865 (2019, Trump Administration) and continued under EO 14110 (2024), CISA is directed to:

- Identify critical infrastructure vulnerable to EMP/CME events
- Develop and disseminate protection guidance
- Coordinate with sector-specific agencies (DOE for energy, DOT for transportation, DOD for defense)
- Conduct EMP/CME exercises and training

2.2 CISA EMP/GMD Guidance Documents

CISA EMP Protection Guidance for Critical Infrastructure (2023)

This guidance document, developed with input from the EMP Commission and national laboratories, recommends: - Layered protection approach (shielding, filtering, isolation) - Priority hardening for assets whose failure would cause cascading consequences - Integration of EMP protection into new construction and major renovations - Annual assessment of EMP protection adequacy

The guidance references but does not mandate dielectric materials specifically. **CSM’s strategy: Work with CISA to include dielectric protection as a recommended best practice in future revisions.**

CISA National Critical Functions Framework (2024 Update)

Identifies 55 National Critical Functions (NCFs) whose disruption would cause national catastrophe. Of these, 31 are electricity-dependent. The framework specifically identifies “Provide Electricity” and “Provide Communications” as Tier 1 functions requiring EMP/CME resilience.

2.3 DHS Grants for EMP Protection

The following DHS grant programs fund critical infrastructure protection:

Program	Annual Funding	EMP Relevance	CSM Opportunity
Homeland Security Grant Program (HSGP)	\$1.12B (FY2025)	State homeland security priorities; EMP hardening eligible	State-level pilot funding
Urban Area Security Initiative (UASI)	\$615M	High-threat urban areas; grid resilience	Urban substation hardening projects
Port Security Grant Program	\$100M	Port electrical infrastructure hardening	Maritime terminal protection
Transit Security Grant Program	\$93M	Electric transit infrastructure	Rail electrification protection
Emergency Management	\$355M	Emergency operations	Civilian C2 facility

SECTION 3: NOAA SPACE WEATHER PREDICTION CENTER

3.1 NOAA SWPC Role

The Space Weather Prediction Center (SWPC) in Boulder, Colorado, is the nation's official source of space weather alerts and warnings. Established in 1965, SWPC operates 24/7 and provides:

- Geomagnetic storm watches and warnings (G-scale, 1-5)
- Solar radiation storm alerts (S-scale, 1-5)
- Radio blackout forecasts (R-scale, 1-5)
- GIC (geomagnetically induced current) modeling for power grid operators

3.2 PROSWIFT Act (2020)

The Promoting Research and Observations of Space Weather to Improve the Forecasting of Tomorrow (PROSWIFT) Act, signed into law in October 2020, directs:

- NOAA to maintain and enhance space weather monitoring capabilities
- DOD to integrate space weather into military planning
- DHS to incorporate space weather into critical infrastructure protection
- Establishment of a Space Weather Advisory Group (SWAG) with industry participation
- Interagency coordination through the Space Weather Operations, Research, and Mitigation (SWORM) working group

Impact on CSM: PROSWIFT institutionalizes space weather as a national security concern, creating sustained funding for monitoring and mitigation. CSM's products are the mitigation component of this framework.

3.3 NOAA Commercial Data Program

NOAA's Commercial Data Program, which purchases weather data from commercial providers, could be extended to space weather. If CSM deploys monitoring equipment as part of hardened installations, this creates a potential data-as-a-service revenue stream.

SECTION 4: EXECUTIVE ORDERS AND NATIONAL SECURITY DIRECTIVES

4.1 Executive Order 13865 (2019) — Coordinating National Resilience to Electromagnetic Pulses

Signed: March 26, 2019 (Trump Administration) Key provisions: - Established EMP as a national security priority - Directed DHS to coordinate EMP resilience across critical infrastructure sectors - Directed DOD to assess EMP vulnerability of military installations - Directed DOE to assess EMP vulnerability of the electric grid - Required annual reporting on EMP preparedness progress

Status: Fully in effect. Annual reports continue through 2026. EO 13865 established the policy framework; subsequent executive orders have expanded scope.

4.2 Executive Order 14110 (2024) — Safe, Secure, and Trustworthy Development and Use of Artificial Intelligence

While primarily about AI, EO 14110 includes a section on critical infrastructure AI applications that references EMP/GMD as a threat to AI-enabled grid control systems. This creates additional regulatory pressure for hardening the control systems that manage modernized, AI-optimized power grids.

4.3 National Security Presidential Memorandum 40 (NSPM-40)

Signed: March 2023 Classification: Unclassified summary published Key provisions: - Designated space weather as a national intelligence priority - Directed integration of space weather into National Intelligence Priorities

Framework - Established space weather threat information sharing between intelligence community and critical infrastructure operators - Created a classified annex on EMP/CME defense priority locations

Impact on CSM: Classified priority location list effectively creates a mandatory hardening requirement for those locations, even if not publicly specified. CSM will need facility security clearances (FCL) to serve these customers.

4.4 EMP Commission Reports

The Congressional EMP Commission (2001-2017, periodically reauthorized) produced the foundational technical assessments that underpin current policy. Its 2008 Critical National Infrastructures Report remains the most-cited source on EMP vulnerability and estimated that a nationwide EMP event would cause:

- 4-10 year recovery timeline for the electric grid
- 65-90% population loss within 12 months due to cascading infrastructure failures (food, water, medicine, transportation)
- \$5-8 trillion in economic damage

The Commission's estimates, while debated, have shaped every federal EMP policy since publication. The Commission was reauthorized in the FY2025 NDAA, with a new report expected in 2027.

SECTION 5: EU CRITICAL ENTITIES RESILIENCE DIRECTIVE

5.1 CER Directive (2022/2557)

Effective: January 2024 (EU member states must transpose by October 2024) Scope: Eleven sectors including energy, transport, banking, financial market infrastructure, health, drinking water, wastewater, digital infrastructure, public administration, space, and food.

Key requirements: - Member states must identify critical entities providing essential services - Critical entities must conduct risk assessments including "natural hazards" - Critical entities must implement resilience measures, including physical protection - Incident reporting to national competent authorities - Cross-border coordination for entities serving multiple member states

EM in the CER Framework: "Natural hazards" as defined in the CER Directive explicitly includes "extreme weather events" and "space weather phenomena" (Recital 18, Annex). This makes EMP/CME protection a compliance requirement for critical entities in the energy and digital infrastructure sectors across all 27 EU member states.

5.2 EU Network Codes on Cybersecurity (Expected 2026-2027)

The EU is developing a Network Code on Cybersecurity for the electricity sector under the 2019 Electricity Regulation. Expected provisions include: - Mandatory risk assessment for electromagnetic threats - Resilience requirements for grid control systems - Certification requirements for protection equipment

Impact on CSM: Creates a 27-country mandatory market for grid hardening products with EU-wide certification standards. CSM should pursue EU certification early to establish first-mover advantage.

5.3 European Space Weather Services

The European Space Agency (ESA) operates the Space Weather Service Network, providing forecasting and alerting for European critical infrastructure operators. ESA's Vigil mission (scheduled launch 2028) will position a spacecraft at the L5 Lagrange point for improved CME detection. This enhanced warning capability will increase demand for mitigation products — knowing a CME is coming creates obligation to protect against it.

SECTION 6: INSURANCE REGULATORY FRAMEWORK

6.1 NAIC (National Association of Insurance Commissioners)

The NAIC coordinates insurance regulation across 50 states, DC, and 5 territories. Key NAIC initiatives relevant to EMP/CME:

NAIC Catastrophe Risk Subgroup - Developing catastrophe risk capital requirements including “emerging perils” - Space weather has been discussed as an emerging peril requiring capital allocation - If NAIC adopts space weather-related risk-based capital charges, insurers would need to either (a) exclude space weather coverage, (b) hold additional capital, or (c) require policyholders to implement hardening measures - CSM’s products enable option (c), which is the lowest-cost path for insurers

NAIC Climate and Resiliency Task Force - 2024 Work Plan includes “assessment of insurance coverage gaps for non-modeled catastrophic perils” - Space weather is the quintessential non-modeled catastrophic peril - Task force is expected to recommend regulatory incentives for resilience investments

NAIC ORSA (Own Risk and Solvency Assessment) - Large insurers must conduct annual ORSA reports identifying material risks - EMP/CME exposure in commercial property portfolios increasingly appears in ORSA filings - Regulators are beginning to question insurers about space weather risk management

6.2 Solvency II (EU)

The EU’s Solvency II insurance regulatory framework requires capital adequacy for all material risks. Key developments:

- **2024 Solvency II Review:** The review process is examining whether “emerging catastrophic risks” including space weather require explicit capital charges
- **EIOPA (European Insurance and Occupational Pensions Authority):** Published a 2024 thematic review on catastrophe risk modeling, identifying space weather as a “significant gap in current modeling”
- If Solvency II imposes space weather capital charges, European insurers will have strong incentive to require EMP/CME hardening for large commercial properties

6.3 IAIS (International Association of Insurance Supervisors)

The IAIS coordinates insurance regulation globally. Its 2024 Global Monitoring Exercise identified “protection gaps for non-modeled catastrophic risks” as a systemic vulnerability. Space weather was specifically referenced.

SECTION 7: REGULATORY TIMELINE — PAST, PRESENT, AND PROJECTED

7.1 Historical Regulatory Milestones

Year	Event	Significance
1965	NOAA SWPC established	Institutional space weather monitoring begins
2001	EMP Commission established by Congress	First comprehensive US EMP vulnerability assessment
2005	Energy Policy Act	FERC/NERC reliability framework established
2008	EMP Commission Critical Infrastructures Report	Definitive EMP threat assessment published
2013	FERC Order 779	GMD standards development initiated
2015	Lloyd’s “Solar Storm Risk” report	First major insurance industry assessment of CME risk
2019	Executive Order 13865	EMP resilience becomes national policy priority
2020	PROSWIFT Act	Space weather monitoring and mitigation codified
2023	NSPM-40	Space weather designated as national intelligence priority
2024	CER Directive effective	EU mandates critical infrastructure resilience including space weather

7.2 Projected Regulatory Milestones

Year	Expected Action	Impact on CSM
2026	NERC GMD Task Force report on protection technologies	Dielectric technology receives formal evaluation
2026 (H2)	NAIC Catastrophe Risk Subgroup space weather discussion paper	Insurance regulatory framework for space weather begins
2027 (H1)	FERC GMD NOPR expected	Mandatory hardening requirements proposed for 4,700+ substations
2027	EMP Commission reauthorized report expected	Updated EMP threat assessment; likely more alarming
2027-2028	EU Network Code on Cybersecurity	EU-wide grid resilience certification requirements
2028	Solvency II amendments effective (2024 review)	Possible space weather capital charges for EU insurers
2028 (H2)	FERC GMD Final Rule expected	Binding mandatory hardening requirements effective 2-5 years after final rule
2028-2029	ESA Vigil mission operational	Improved CME warning creates mitigation demand
2029-2030	IMO space weather guidelines for maritime	Maritime vessel EMP protection standards
2030	FERC GMD Final Rule implementation deadline (estimated)	Peak demand for substation hardening equipment
2032	CER Directive review cycle	Possible expansion of “critical entities” definition

7.3 CSM Regulatory Engagement Strategy

Immediate Actions (2026-2027)

1. **FERC/NERC engagement:** Participate in NERC GMD Task Force; submit technical comments on protection technology performance requirements. Position dielectric technology as preferred compliance pathway.
2. **CISA relationship:** Establish CISA engagement through the Critical Infrastructure Partnership Advisory Council (CIPAC) framework. Offer CSM expertise for EMP exercise design.
3. **Congressional engagement:** Brief relevant committees (House Energy & Commerce, Senate Energy & Natural Resources, House Homeland Security) on dielectric protection technology. Advocate for technology-neutral hardening standards that include dielectric approaches.
4. **NAIC engagement:** Participate in Catastrophe Risk Subgroup proceedings. Provide technical input on space weather risk assessment and mitigation. Encourage premium credit requirements for hardened assets.
5. **EU engagement:** Establish Brussels presence for CER Directive implementation engagement. Participate in ENTSO-E (European Network of Transmission System Operators) GMD working groups.
6. **Standards development:** Initiate IEEE/IEC standards working group for dielectric EMP protection performance testing and certification. Standards create market acceptance and regulatory referenceability.

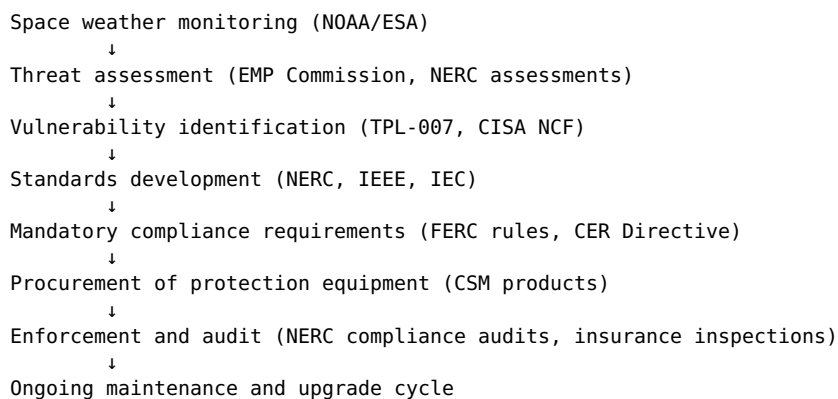
Medium-Term Actions (2028-2030)

7. Obtain FCL (Facility Clearance) for classified government work. Register with System for Award Management (SAM.gov) and obtain CAGE code for government contracting.
8. Participate in EMP exercise programs (DOD’s “Sudden Dawn” exercise series, DHS National Level Exercise program) to demonstrate technology in operational scenarios.
9. Advocate for inclusion of EMP/CME protection in building codes (ICC, NFPA 70 National Electrical Code) for critical facilities.

SECTION 8: HOW REGULATIONS CREATE MANDATORY DEMAND

8.1 The Regulatory Demand Cascade

Regulatory demand for EMP/CME protection follows a predictable cascade:



CSM is positioned at step 5 (procurement). The preceding steps are occurring now and will produce mandatory, non-discretionary demand by 2028-2030.

8.2 Regulatory Demand vs. Economic Demand

Unlike products that depend on customer ROI calculations, mandated products have demand curves that are: - **Inelastic**: Demand does not decrease if prices rise (within reason) - **Non-cyclical**: Not affected by economic downturns - **Legislated**: Timing determined by regulatory calendars, not market conditions - **Penalty-backed**: Non-compliance carries fines, operating restrictions, or liability exposure

This regulatory demand profile is attractive to investors because it reduces revenue uncertainty and competitive pricing pressure.

8.3 Director and Officer Liability

A secondary demand driver is D&O (director and officer) liability. Public company boards that fail to address known catastrophic risks may face shareholder derivative lawsuits. The 2023 Caremark standard (Delaware Chancery Court) requires boards to exercise oversight of “mission-critical” risks. As EMP/CME risk becomes better understood, utility board members and officers face increasing personal liability exposure for failing to implement available protective measures. CSM’s products are the most direct way for utilities to demonstrate reasonable oversight of EMP/CME risk.

SECTION 9: RISK FACTORS

1. **Regulatory delay**: The FERC GMD NOPR could be delayed by political changes or industry opposition. Each year of delay shifts CSM’s revenue curve right by 1-2 years.
2. **Regulatory capture**: Large incumbent utilities and grid equipment manufacturers may influence standards to favor their existing products or create compliance pathways that don’t require new technology.
3. **Preemption risk**: Federal grid resilience standards could preempt more stringent state standards, reducing overall demand.
4. **Technology specification risk**: If NERC standards specify particular technologies (e.g., “neutral blocking devices” vs. “GIC mitigation”), they could exclude dielectric approaches from compliance.
5. **International divergence**: If US and EU standards diverge, CSM would need separate product lines for each market, increasing development cost.
6. **Enforcement weakness**: Standards without enforcement are suggestions. If NERC compliance audits are underfunded, mandated demand may not materialize.

7. **Insurance market withdrawal:** If insurers simply exclude EMP/CME from coverage rather than requiring hardening, the insurance distribution channel disappears.

Document prepared by: Carrington Storm Motors Regulatory Affairs Data sources: FERC, NERC, DHS/CISA, NOAA SWPC, EMP Commission, NAIC, EIOPA, IAIS, European Commission, US Code and Executive Orders Last updated: July 15, 2026